

Market Reaction and Volatility Dynamics Around Major Technology Conferences

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Abstract

This project investigates how equity markets react to major technology conferences and whether certain events consistently generate abnormal returns and volatility. Using historical stock price data from Apple, Nvidia, Google, Meta, and Microsoft, an event-study framework is applied to evaluate abnormal returns, cumulative abnormal returns, volatility changes, and distributional characteristics of market reactions. The study further compares AI-focused conferences with traditional technology conferences to determine whether artificial intelligence announcements generate stronger market responses.

1 Introduction

Corporate technology conferences often serve as important information events where firms announce new products, technological developments, artificial intelligence capabilities, strategic initiatives, and future growth plans. Financial markets rapidly process this information and incorporate it into stock prices.

The objective of this study is to investigate whether major technology conferences produce abnormal stock returns and whether AI-focused events generate stronger market reactions than traditional technology conferences.

The primary research questions are:

1. Which technology conferences generate the largest stock price reactions?
2. Do these events produce abnormal returns relative to the broader market?
3. Are AI-focused conferences associated with stronger market reactions than traditional technology conferences?
4. Does pre-event volatility predict the magnitude of event-day movements?

2 Data

Historical stock price data was collected using Yahoo Finance through the Python package `yfinance`. The sample period covers major technology conference events between 2018 and 2025.

The following companies and conferences were analyzed:

Company	Ticker	Conference
Apple	AAPL	WWDC
Nvidia	NVDA	GTC
Google	GOOGL	Google I/O
Meta	META	Meta Connect
Microsoft	MSFT	Build

Table 1: Companies and Conferences Included in the Study

The S&P 500 ETF, SPY, was used as the market benchmark. SPY was selected because it provides a broad representation of the U.S. equity market and allows each stock’s event return to be compared against general market movement.

3 Methodology

3.1 Stock Return

The one-day stock return around each conference was calculated as:

$$R = \frac{P_{after} - P_{before}}{P_{before}}$$

where:

- P_{before} is the closing price before the conference.
- P_{after} is the closing price after the conference.

3.2 Abnormal Return

Abnormal return measures the stock’s performance relative to the broader market:

$$AR = R_{stock} - R_{market}$$

where R_{market} is represented by SPY’s return over the same event window.

A positive abnormal return indicates that the stock outperformed the market around the conference. A negative abnormal return indicates that the stock underperformed the market.

3.3 Cumulative Abnormal Return

To capture delayed investor reactions, a three-day event window was also used. Cumulative abnormal return was calculated as:

$$CAR = \sum AR_t$$

This metric measures the total abnormal return generated over the event window.

3.4 Absolute Move

The absolute stock move was used to measure the magnitude of the market reaction regardless of direction:

$$\text{Absolute Move} = |R_{stock}|$$

This helps identify which conferences generated the largest price reactions.

3.5 Volatility

Realized volatility was calculated using a 20-trading-day window before and after each conference. Annualized volatility was estimated as:

$$\sigma = std(r)\sqrt{252}$$

where:

- r represents daily returns.
- 252 represents the approximate number of trading days in a year.

3.6 Distribution Analysis

The distribution of abnormal returns was analyzed using:

- Mean
- Standard deviation
- Skewness
- Kurtosis

Skewness measures whether returns are tilted toward positive or negative surprises. Kurtosis measures the extent of tail behavior in the return distribution.

3.7 Hypothesis Testing

A one-sample t-test was conducted to determine whether the average abnormal return was statistically different from zero:

$$H_0 : AR = 0$$

$$H_1 : AR \neq 0$$

The null hypothesis assumes that the conference does not generate abnormal returns.

4 Results

4.1 Average Absolute Stock Movement

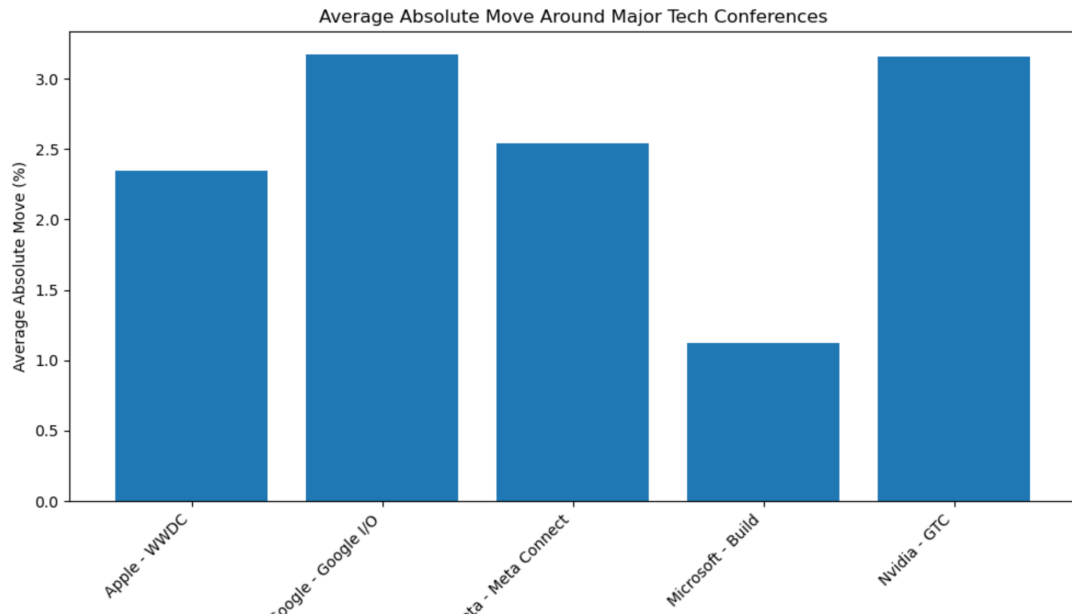


Figure 1: Average absolute stock price movement around major technology conferences. Google I/O and Nvidia GTC generated the largest average price reactions, while Microsoft Build produced the smallest reaction.

Google I/O and Nvidia GTC produced the largest average absolute price movements, both exceeding 3%. Apple WWDC and Meta Connect generated moderate price reactions, while Microsoft Build generated the smallest average absolute move.

4.2 Conference Ranking by Average Abnormal Return

Conference	Mean Abnormal Return (%)
Google I/O	2.32
Nvidia GTC	1.67
Apple WWDC	1.21
Microsoft Build	0.43
Meta Connect	-0.10

Table 2: Average Abnormal Return Ranking

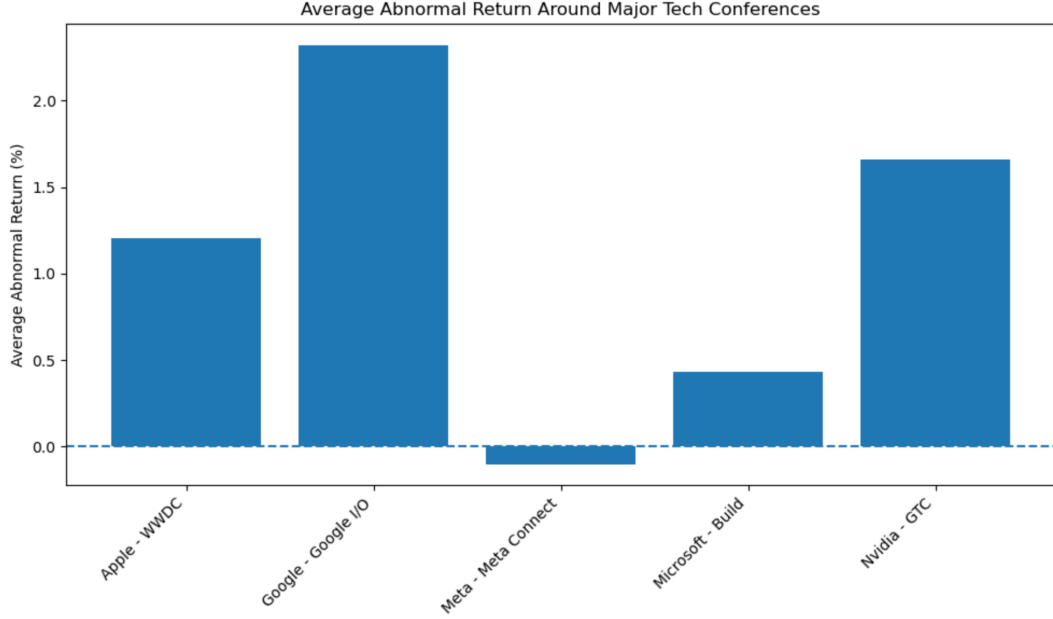


Figure 2: Average abnormal returns around major technology conferences. Google I/O generated the highest abnormal return, followed by Nvidia GTC and Apple WWDC. Meta Connect produced a slightly negative abnormal return.

Google I/O generated the strongest average abnormal return, followed by Nvidia GTC and Apple WWDC. This suggests that investors responded most positively to Google and Nvidia conference announcements relative to overall market movement.

Meta Connect produced a slightly negative average abnormal return, indicating that Meta’s conference announcements did not generate consistent positive market reactions during the sample period.

4.3 Cumulative Abnormal Return

Conference	Mean CAR 3-Day (%)
Nvidia GTC	4.12
Google I/O	2.66
Apple WWDC	1.41
Microsoft Build	1.04
Meta Connect	-0.50

Table 3: Three-Day Cumulative Abnormal Returns

Nvidia GTC produced the strongest cumulative abnormal return over the three-day event window. This suggests that investors continued reacting positively after Nvidia’s conference announcements, possibly because of delayed interpretation of artificial intelligence and semiconductor-related news.

Google I/O also generated a strong positive cumulative abnormal return. Apple WWDC and Microsoft Build generated positive but smaller cumulative abnormal returns, while Meta Connect generated a negative average cumulative abnormal return.

4.4 Distribution of Abnormal Returns

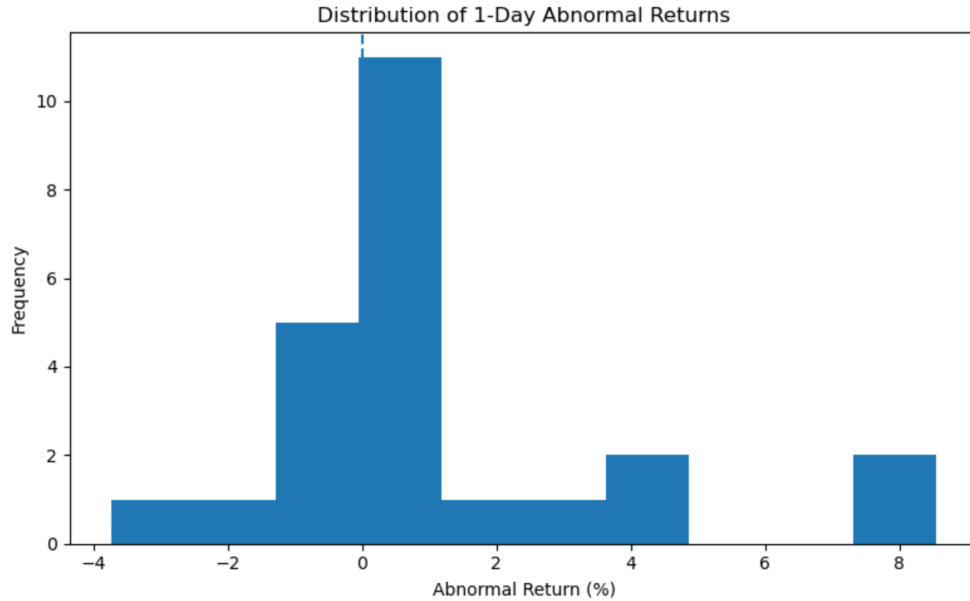


Figure 3: Distribution of one-day abnormal returns across all conference events. Most observations cluster around zero, with a small number of large positive reactions creating a right-skewed distribution.

Most abnormal returns were concentrated around zero, meaning that many conference events produced modest market reactions. However, a few large positive abnormal returns created a right-skewed distribution. This supports the idea that some technology conferences generate meaningful upside surprises.

4.5 Skewness and Kurtosis

Conference	Skewness	Kurtosis
Apple WWDC	0.54	-0.79
Google I/O	1.15	-0.67
Meta Connect	-0.61	-1.03
Microsoft Build	-0.11	-1.11
Nvidia GTC	1.02	-0.78

Table 4: Skewness and Kurtosis of Abnormal Returns

Google I/O and Nvidia GTC exhibited strong positive skewness. This suggests that when large surprises occurred around these conferences, they were more likely to be positive than negative.

Meta Connect displayed negative skewness, indicating weaker or more negative surprise behavior during the sample period.

All events exhibited negative kurtosis, suggesting fewer extreme observations than would be expected under a normal distribution. However, this result should be interpreted cautiously because the sample size is small.

4.6 Hypothesis Testing

Conference	t-statistic	p-value
Apple WWDC	1.66	0.14
Google I/O	1.16	0.33
Meta Connect	-0.08	0.94
Microsoft Build	1.22	0.31
Nvidia GTC	0.71	0.53

Table 5: One-Sample t-Test Results for Abnormal Returns

All p-values exceed the conventional 5% significance threshold. Therefore, the null hypothesis that average abnormal returns equal zero cannot be rejected.

This does not necessarily mean that conferences have no market impact. Rather, the statistical test is limited by the small number of available observations, since these conferences occur only once per year.

4.7 AI-Focused vs Traditional Technology Conferences

To investigate whether AI-related conferences generate stronger market reactions than traditional technology conferences, events were classified into two groups.

Nvidia GTC and Google I/O were categorized as AI-focused conferences due to their emphasis on artificial intelligence, machine learning, cloud computing, and semiconductor innovation. Apple WWDC, Microsoft Build, and Meta Connect were classified as traditional technology conferences.

Category	Avg Move (%)	Avg AR (%)	Avg CAR (%)
AI-Focused	3.16	1.99	3.39
Traditional Technology	2.09	0.69	0.84

Table 6: AI-Focused vs Traditional Technology Conferences

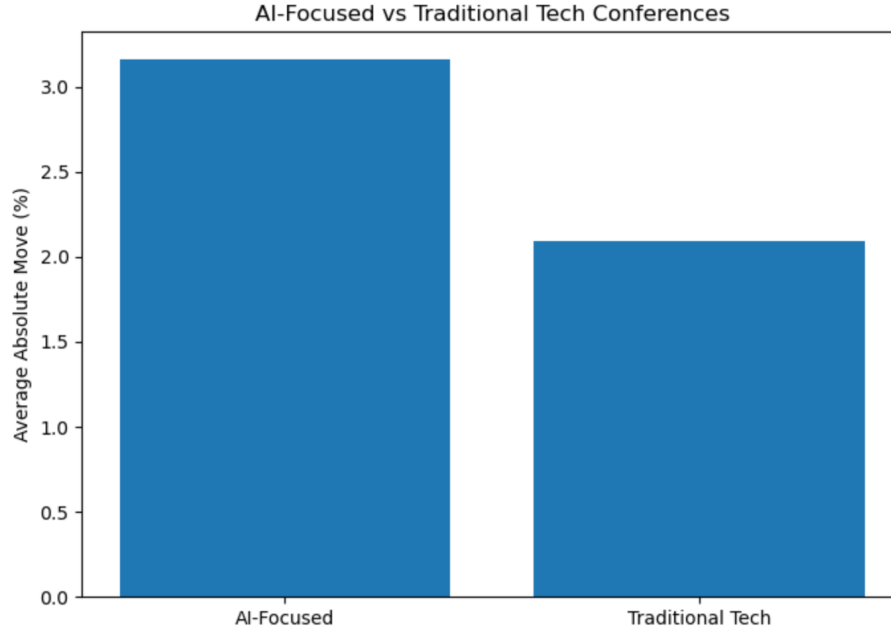


Figure 4: Comparison of AI-focused and traditional technology conferences. AI-focused events generated larger average stock price reactions than traditional technology conferences.

AI-focused conferences generated larger average absolute price movements, abnormal returns, and cumulative abnormal returns than traditional technology conferences.

This suggests that investors currently place greater value on AI-related announcements and innovation than on broader technology updates.

4.8 Volatility Prediction Analysis

The final part of the analysis investigated whether pre-event volatility helps predict the magnitude of conference-day stock movement.

The correlation between pre-event volatility and event-day movement was:

$$\rho = -0.13$$

A simple linear regression was also estimated, where pre-event volatility was the independent variable and the absolute event-day move was the dependent variable.

The regression results were:

- Slope = -0.0187
- Intercept = 3.1066
- $R^2 = 0.017$
- p-value = 0.54

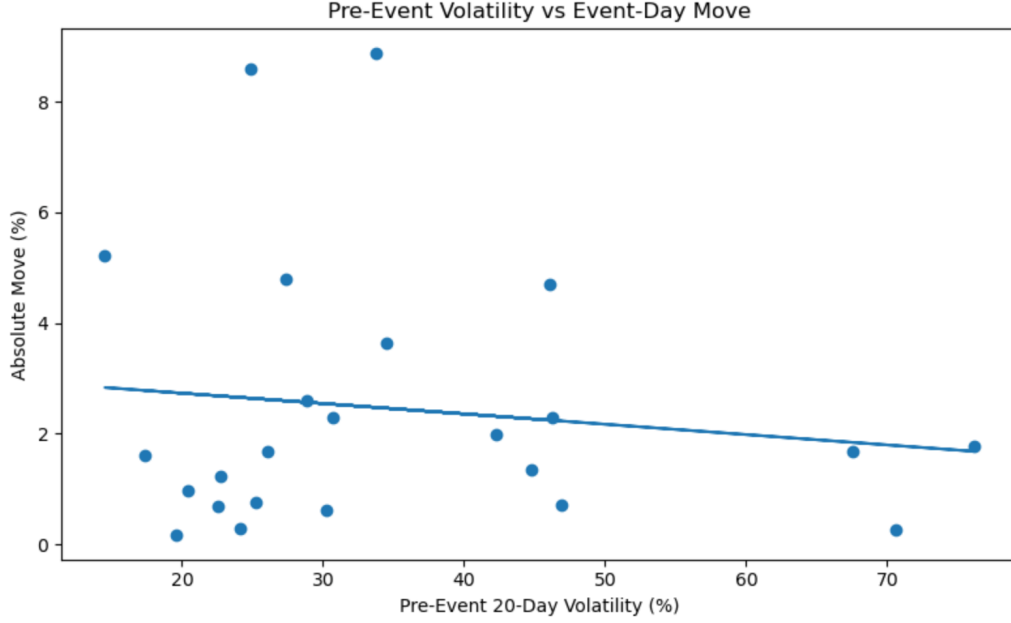


Figure 5: Relationship between pre-event volatility and event-day stock movement. The fitted regression line suggests little predictive relationship between pre-event volatility and conference-day price reactions.

The low R^2 value indicates that pre-event volatility explains only about 1.7% of the variation in event-day stock movements. The p-value of 0.54 indicates that the relationship is not statistically significant.

Therefore, pre-event volatility did not provide meaningful predictive power for the magnitude of conference-day stock reactions.

5 Key Findings

The main findings of the project are:

1. Google I/O generated the highest average abnormal return among the conferences studied.
2. Nvidia GTC produced the strongest three-day cumulative abnormal return.
3. AI-focused conferences generated larger market reactions than traditional technology conferences.
4. Google I/O and Nvidia GTC exhibited positive skewness, suggesting that large surprises were more likely to be favorable.
5. Pre-event volatility was not a reliable predictor of event-day stock price movement.
6. Statistical significance was limited due to the small number of conference observations.

6 Limitations

Several limitations should be considered when interpreting the results.

6.1 Small Sample Size

Technology conferences usually occur once per year. As a result, the number of observations is limited. This reduces the statistical power of hypothesis tests and makes it harder to draw strong conclusions.

6.2 Use of Free Market Data

The project used Yahoo Finance data through `yfinance`. While useful and accessible, this data may not match institutional-grade datasets in terms of precision, historical adjustments, or completeness.

6.3 Daily Closing Prices Only

The analysis uses daily closing prices. Intraday data could provide a more precise view of how quickly markets react to conference announcements.

6.4 No Options Market Data

The original version of this project considered using ATM straddle prices to estimate option-implied moves around major technology conferences. However, historical option chain data is not freely available through Yahoo Finance and usually requires paid institutional databases such as OptionMetrics, CBOE DataShop, Polygon.io, or WRDS access.

Therefore, this project was redesigned as a free-data event study using equity prices and SPY benchmark returns. The revised approach still captures market reaction, abnormal returns, realized volatility changes, skewness, kurtosis, and event-driven risk behavior while remaining fully reproducible with public data.

6.5 External Market Factors

Conference-day returns may be affected by macroeconomic news, earnings expectations, interest rate announcements, or broader technology-sector trends. Although SPY was used as a benchmark, not all external influences can be fully removed.

7 Conclusion

This study demonstrates that major technology conferences can generate meaningful market reactions, particularly when associated with artificial intelligence developments.

Among the events analyzed, Google I/O generated the highest average abnormal return, while Nvidia GTC produced the strongest cumulative abnormal return over a three-day event window. AI-focused conferences consistently produced larger absolute price movements, abnormal returns, and cumulative abnormal returns than traditional technology conferences.

However, the results should be interpreted cautiously due to the limited sample size and lack of statistically significant t-test results. The analysis shows evidence of economically meaningful patterns, but stronger conclusions would require a larger dataset or more granular intraday and options market data.

Overall, this project provides a practical application of event-study methodology in quantitative finance and demonstrates how publicly available data can be used to analyze market reactions to corporate innovation events.

Future extensions could include options-implied volatility, intraday event windows, sentiment analysis of keynote announcements, earnings controls, and a broader universe of technology firms.